

Tachycardia (Wide Complex with Pulses)

History

- Is the patient tachycardic (HR > 100)?
- Is the patient symptomatic?
- If symptomatic, when did the symptoms start?
- Did patient have recent trauma? Toxic ingestion?
- Does patient have suspected infection? Fever? Heat exposure? Dehydration?
- Patient's medical history – Cardiac history? History of arrhythmia? History of prolonged QT syndrome? Renal failure? Diabetes? Lung disease?
- Medications – Many medications can lead to tachyarrhythmias

Signs and Symptoms of Hemodynamically Significant Tachycardia

- Hypotension
- Syncope / Dizziness / lightheadedness
- Chest pain
- Shortness of breath / Difficulty breathing
- Signs of pulmonary edema (crackles / rales, hypoxia)
- Altered mental status / Agitation
- Diaphoresis
- Cyanosis or decreased capillary refill

EMT/AEMT

- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline, as indicated

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO bolus for SBP < 90 or signs/symptoms of shock

Paramedic

- Advanced airway as indicated
- 12 – Lead EKG
- **NS/LR** 500-1000mL IV/IO bolus for SBP ≤ 90 or signs/symptoms of shock

Sinus Tachycardia or Heart Rate < 150

- 12 – Lead interpretation

Heart Rate > 150 and SBP ≤ 90

- Immediate **Synchronized Cardioversion** – 100 J → 150 J → 200 J biphasic
 - Reassess rhythm between cardioversions
 - If no cardioversion after 200J
 - Consult Medical Direction for further orders
 - If patient converts
 - Repeat 12 – Lead
 - If SBP > 90, **Amiodarone** 150 mg IV/IO over 10 minutes, (**avoid or stop if SBP < 90**)

Heart Rate > 150 and SBP > 90 with Regular Rhythm (VTach or uncertain)

- Monomorphic Regular Tachycardia give **Adenosine** 12 mg IV
- *If Adenosine not effective* give **Amiodarone** 150 mg IV/IO over 10 minutes
 - If SBP < 90, stop Amiodarone and consider **Synchronized Cardioversion**
 - If conversion, repeat 12 – Lead and repeat vital signs
- If **no** conversion consider additional **Amiodarone** 150 mg IV/IO over 10 minutes

Heart Rate > 150 and SBP > 90 with Irregular Rhythm (a-fib, a-flutter with aberrancy)

- **Amiodarone** 150 mg IV/IO over 10 minutes
 - If SBP < 90, stop Amiodarone and consider **Synchronized Cardioversion**
- **Magnesium sulfate** 1 g IV/IO over 10 minutes, if Torsades de Pointes